



Advanced Card Systems Ltd.
Identification and Payment Technologies



Identification & Payment Technologies

PRODUCT CATALOGUE



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- AdvancedCardSystems
- AdvancedCardSystems
- Advanced Card Systems Ltd.
- advancedcardsystems

2026

CORPORATE PROFILE

Advanced Card Systems Ltd. (ACS, wholly owned subsidiary of SEHK: 2086), founded in 1995, is Asia Pacific's top supplier and one of the world's top 3 suppliers of PC-linked smart card readers, as well as the winner of the Product Quality Leadership Award for Smart Card Readers from Frost & Sullivan. ACS was listed in Forbes Asia's "Best Under a Billion" list, an inter-industry list comprised of 200 top-performing publicly listed companies in Asia Pacific, with sales between US\$5 million and US\$1 billion for several years. ACS develops a wide range of high-quality smart card reading/writing devices, smart cards and related products and distributes them to over 100 countries worldwide.

Vision

Combine scientific spirit and Confucius thoughts to build a sustainable electronic and IT business that is green in order to achieve the dual purposes of business growth and environmental protection.

Mission

To strengthen its leading position as a provider of identification and payment technologies in the world.



Asia Pacific's No. 1 and Among The World's Top 3 Suppliers of PC-Linked Smart Card Readers



20+ International & Industry-Level Awards and Recognitions



200M+ Global Users



More than 50% Workforce Comprised of R & D



200+ High-Quality Products Developed



100+ Countries Served



30K+ Ticket Validators Deployed



50+ Cities Benefitting from our Contactless Payment Technologies

PRODUCTS



Smart Card Readers

ACS develops and delivers high-quality, reliable smart card reader solutions across contact, contactless, and embeddable form factors. Our contact readers adhere to industry standards such as PC/SC and EMV, while our contactless readers/writers meet the growing NFC demand for e-government, e-banking, and other tap-and-go applications. For OEM integration, ACS provides compact modules with optimized RF performance, ideal for kiosks, machines, and device embedding.



Contact Card Readers / Contactless Card Readers / Card Reader Modules



Smart Cards & Smart Card Operating Systems

ACS develops smart card with a proprietary smart card operation system (ACOS). ACOS cards meet the security requirements of these applications via multilevel secured access hierarchy.



ID Document Readers

ACS brand new product line ID Document Readers deliver compact, versatile identity verification for space-constrained counters and kiosks. A modular design unifies OCR, RFID/NFC, and contact smart card reading to capture data quickly and reliably from ICAO Doc 9303-compliant e-passports and ID cards.



FIDO® Device

ACS's FIDO® certified devices are compliant with FIDO U2F and FIDO2 protocols, and aim to provide users with a speedy and safeguarded passwordless online login experience.



Automatic Fare Collection (AFC) Readers

ACS designs and delivers multi-interface devices powering large-scale transportation ecosystems—urban transit, intercity rail, and parking networks to ensure fast, secure fare validation and payments.



SINCE 1995



SMART CARD READERS – CONTACT

ACS develops and provides high-quality and reliable smart card readers with contact interface, based on various industry standards such as PC/SC (Personal Computer/Smart Card) and EMV (Europay, MasterCard and Visa). ACS contact smart card readers are ideal for individual authentication in security applications such as e-government services, e-commerce, banking services, healthcare management, digital signature and internet lottery.

						
ACR40U-A1 / ACR40U-AF	ACR40U-B1 / ACR40U-BF	ACR40T-A1 / ACR40T-A5	ACR40T-A6 / ACR40T-A7	ACR39U-N1 / ACR39U-NF	ACR3901U-S1	APG8201-B2

Physical Characteristics															
Dimensions (mm)	55,0 × 65,0 × 9,0		58,5 × 67,6 × 62,5		48,9 × 17,7 × 8,0 (A1) 44,7 × 17,7 × 8,0 (A5)		49,5 × 17,7 × 8,0 (A6) 45,9 × 17,7 × 8,0 (A7)		58,0 × 20,0 × 13,9		94,0 X 60,0 X 12,0		96,0 X 60,0 X 13,0		
Weight (g)	40,0		65,0		6,8 (A1) / 5,3 (A5)		8,1 (A6) / 6,2 (A7)		12,0 (N1) / 10,0 (NF)		30,8		98,0		
Host Interface															
USB (Full Speed)		 		 		 		 		 					
Bluetooth® Standard												Bluetooth 4,0			
Contact Smart Card Interface															
ISO 7816 (Class A, B and C)	●		●		●		●		●		●		●		
MCU Cards, T=0 and T=1	●		●		●		●		●		●		●		
Memory Cards	●		●		●		●		●		●		●		
Smart Card Read/Write Speed (kbps)	600		600		600		600		600		600		250		
Card Form	Full-Sized		Full-Sized		SIM-Sized		SIM-Sized		Full-Sized		Full-Sized		Full-Sized		
Upgradeability															
Firmware Upgradeable	●		●		●		●				●		●		
Built-In Peripherals															
LED	2		2		1		1		1		3				
LCD													●		
Keypad													20-Key		
Buzzer											●				
Button							●						●		
Certifications/Compliance															
PC/SC, CCID, WHQL	●		●		●		●		●		●		●		
EMV Level 1 (Contact)	●		●						●						
CE, FCC, UKCA	●		●		●		●		●		●		●		
KC (Korea)	●		●						●						
RoHS, REACH	●		●		●		●		●		●		●		
Bluetooth	●		●						●		●				
J-LIS (Japan)	●		●						●		●				
VCCI (Japan)	●		●		●		●		●		●				
MIC (Japan)											●				
MTBF (hours)	500,000		500,000		500,000		500,000		500,000		400,000		60,000		
Supported OS Platforms															
Windows	●		●		●		●		●		●		●		
macOS	●		●		●		●		●		PC-Linked Mode		●		
Linux	●		●		●		●		●		PC-Linked Mode		●		
Android	●		●		●		●		●		●		●		
iOS/ iPadOS (iOS16 or above)	●		●		●		●		●		●		●		

Subject to change without prior notice



Full-Sized



Sim-Sized



SMART CARD READERS – CONTACTLESS

ACS offers contactless smart card readers/writers to address the growing demand for contactless applications. Hinged upon the 13.56 MHz RFID technology, these readers are developed to support various contactless protocols, such as ISO 14443 Type A and B, ISO18092, ISO15693, MiFare®, FeliCa and NFC, to facilitate their use in a wide range of applications. Additionally, ACS provides dual interface readers that allow for the integration of both contact and contactless technologies into a single device and card, consolidating traditionally separate applications.



Physical Characteristics						
Dimensions (mm)	98,0 x 65,0 x 12,8	93,0 x 58,0 x 6,0	85,0 x 60,0 x 9,0	120,5 x 72,0 x 20,4	75,0 x 30,0 x 12,7	158,0 x 95,0 x 95,0 (w/ Base Stand)
Weight (g)	89,0	31,0	67,0	143,0	15,8	415,0 (w/ Base Stand)
Processor						
Processor	32 bit	32 bit	32 bit	32 bit	8 bit	8 bit
Power						
USB Powered	●	●	●	●	●	●
Battery Powered		1 x Rechargeable Li-ion Battery (charging through USB)				
Host Interface						
USB (Full Speed)	 ACR1552U-M1 USB Type-A	 ACR1552U-MF USB Type-C	 ACR1555U-A1 USB Type-C to Type-A	 ACR1552U-A2 USB Type-A	 ACR1581U-C1 USB Type-A	 ACR1581U-CF USB Type-C
					 ACR1251T USB Type-A	 ACR1222L USB Type-A
Bluetooth® Standard		BT 5.2 BLE (2 Mbps)				
Contactless Smart Card Interface						
ISO 14443 Type A and B	●	●	●	●	●	●
ISO/IEC 18092 (NFC)	●	●	●	●	●	●
ISO 15693	●	●	●	●	●	●
MI FARE®	●	●	●	●	●	●
FeliCa	●	●	●	●	●	●
Smart Card Read/Write Speed (kbps)	106, 212, 424, 848	106, 212, 424, 848	106, 212, 424, 848	106, 212, 424, 848	106, 212, 424	106, 212, 424
Reading Distance (mm) (depends on tag type)	70	50	70	70	30	50
Extended APDU Support	●	●	●	●	●	●
NFC Reader/Writer Mode	●	●	●	●	●	●
NFC Card Emulation Mode	●	●	●	●	●	●
NFC Keyboard Emulation Mode	●	●	●	●	●	●
Contact Smart Card Interface						
ISO 7816 (Class A, B and C)				●		
MCU Cards, T=0 and T=1				●		
Smart Card Read/Write Speed (kbps)				1250		
Upgradeability						
USB Firmware Upgrade	●	●	●	●	●	●
Built-In Peripherals						
LCD						●
LED	1 bi-colour	1 bi-colour + 2	1 RGB colour	2	1 bi-colour	4
Buzzer	●	●	●	●	●	●
SAM Slot	1	1		1		3
Certifications/Compliance						
PC/SC, CCID, WHQL	●	●	●	●	●	●
CE, FCC, UKCA	●	●	●	●	●	●
J-LIS (Japan)	●	●	●	●	●	●
VCCI, MIC (Japan)	●	●	●	●	●	●
TELEC (Japan)	●	●	●	●	●	●
RoHS, REACH	●	●	●	●	●	●
Bluetooth		●			●	●
IP Rating			IP67			
Supported OS Platforms						
Windows	●	●	●	●	●	●
macOS	●	PC-Linked Mode	●	●	●	●
Linux	●	PC-Linked Mode	●	●	●	●
Android	●	●	●	●	●	●
iOS/iPadOS (iOS16 or above)	●	●	●	●	●	●

Subject to change without prior notice



Extensive Card Compatibility
Support ISO 14443A/B, ISO18092, ISO15693



High Read/Write Speed
Up to 848kbps



High Reading Distance
Up to 7cm





WALLETMATE SERIES

The WalletMate series Mobile Wallet NFC Reader and modules enable seamless mobile NFC applications. Certified for Apple VAS (ECP 1.0), Apple Access (ECP 2.0) and Google Smart Tap, it supports virtual passes such as loyalty cards, tickets, and IDs on iOS and Android, and delivers secure, fast, and efficient transactions.



Physical Characteristics			
Dimensions (mm)	98.0 x 65.0 x 12.8	57.0 x 20.0 x 5.7	106.6 x 67.0 x 16.0
Weight (g)	89.0	3.9	20.8
Processor			
Processor	32 bit	32 bit	32 bit
Host Interface			
USB (Full Speed)	 WalletMate II USB Type-A	5pin, 1mm pitch, JST SH series compatible	2.54mm Pitch
Serial (RS-232)	2.54mm Pitch & 2.5mm Pitch		
Contactless Smart Card Interface			
ISO 14443 Type A and B	●	●	●
ISO/IEC 18092 (NFC)	●	●	●
ISO15693	●	●	●
MIFARE®	●	●	●
FeliCa	●	●	●
Smart Card Read/Write Speed (kbps)	106, 212, 424, 848	106, 212, 424, 848	106, 212, 424, 848
Reading Distance (mm) (depends on tag type)	Up to 70	Up to 50	Up to 70
Extended APDU Support	●	●	●
NFC Reader/Writer Mode	●	●	●
NFC Card Emulation Mode	●	●	●
NFC Keyboard Emulation Mode	●	●	●
Upgradeability			
USB Firmware Upgrade	●	●	●
Built-In Peripherals			
LED	1 bi-colour	1 bi-colour	2
Buzzer	●		●
SAM Slot	1		1
Certifications/Compliance			
PC/SC, CCID, WHQL	●	●	●
Apple VAS (ECP 1.0), Apple Access (ECP 2.0)	●	●	●
Google Smart Tap	●	●	●
CE, FCC, UKCA	●	●	●
J-LIS (Japan)	●	●	●
VCCI, MIC (Japan)	●	●	●
RoHS, REACH	●	●	●
Supported OS Platforms			
Windows	●	●	●
macOS	●	●	●
Linux	●	●	●
Android	●	●	●
iOS/iPad OS (iOS16 or above)	●	●	●

Subject to change without prior notice



SMART CARD READER MODULES

ACS develops smart card reader solutions in embeddable form. These smart card & contactless reader modules are designed to be integrated into embedded systems, which include smart card readers, ATM machines, kiosks, and gaming machines. Support various contactless protocols, such as ISO18092 and ISO15693. SAM slots add layers of security to the modules. This product line offers both contact and contactless reader module options to help it add value to a wide range of applications.



Physical Characteristics			
Dimensions (mm)	55.0 x 45.0 x 6.5 (Main Board) 91.0 x 49.5 x 5.0 (Antenna Board)	52.0 x 20.0 x 5.7	106.6 x 67.0 x 16.0
Weight (g)	27.7	3.4	20.8
Host Interface			
USB (Full Speed)	●	●	●
Serial (RS-232)			●
Connector Type	5 pin, 1.25mm pitch, Molex PicoBlade series compatible	5pin, 1mm pitch, JST SH series compatible	2.54mm Pitch (USB) & 2.5mm Pitch (RS232)
Contactless Smart Card Interface			
ISO 14443 Type A and B	●	●	●
ISO/IEC 18092 (NFC)	●	●	●
ISO 15693	●	●	●
MIFARE®	●	●	●
FeliCa	●	●	●
Smart Card Read/Write Speed (kbps)	106, 212, 424, 848	106, 212, 424, 848	106, 212, 424, 848
Reading Distance (mm)	Up to 70	Up to 50	Up to 70
Extended APDU Support	●	●	●
NFC Reader/Writer Mode	●	●	●
NFC Card Emulation Mode	●	●	●
NFC Keyboard Emulation Mode	●	●	●
Upgradeability			
Firmware Upgradable	●	●	●
Built-In Peripherals			
SAM Slot	Upon Request		1
LED	1 bi-colour	1 bi-colour	2
Buzzer	●		●
Certifications/Compliance			
PC/SC, CCID, WHQL	●	●	●
CE, FCC	●	●	●
VCCI (Japan)	●	●	●
MIC (Japan)	●	●	●
RoHS, REACH	●	●	●
Supported OS Platforms			
Windows	●	●	●
macOS	●	●	●
Linux	●	●	●
Android	●	●	●
iOS/iPadOS (iOS16 or above)	●	●	●

Subject to change without prior notice





SMART CARDS & SMART CARD OPERATING SYSTEMS

ACS develops and provides smart cards with a proprietary smart card operating system (ACOS). Often noted for their ‘secure and elegant’ architecture, ACOS cards uncover the powerful potential of smart cards, enabling a single card to support multiple applications, from physical/network access control to payment. ACOS cards meet the security requirements of these applications via multilevel secured access hierarchy.

One of these cards, the ACOS5-EVO, is the latest addition to the ACOS5 Series, that provides additional features such as support for ECC and RSA algorithms and extended APDU.



SMART CARD
Robust ACOS Architecture

CRYPTOGRAPHIC USB TOKEN
Portable PKI Tokens



				
ACOS3	ACOS5-K1 (ACOS5-EVO)	ACOS5T2-B1E1 (CryptoMate EVO)	ACOSJ	ACOSJ-GQ

Form Factor					
Card - Contact Only		•	•	•	•
Card - Contactless Only		•	•	•	•
Card - Combi (Contact and Contactless)		•	•	•	•
Token - Contact Card and USB Full Speed			•		
Communication Speed					
Contact Interface		up to 223.2 kbps	up to 446.4 kbps	up to 446.4 kbps	up to 625 kbps
Contactless Interface		up to 848 kbps	up to 424 kbps	up to 848 kbps	up to 848 kbps
Memory (EEPROM Size)					
Contact Interface		32 KB/ 72 KB/ 256 KB	192 KB	192 KB	95 KB
Contactless Interface		8 KB	192 KB	192 KB	95 KB
Combi Interface		8 KB	192 KB	192 KB	95 KB
Protocol					
Contact Interface	T = 0	•	•	•	•
	T = 1		• (Default)	• (Default)	•
Contactless Interface	T = CL	•	•	•	•
Certifications / Compliance					
Contact Interface	ISO 7816 - 1/2/3	•	•	•	•
	ISO 7816 - 4		•	•	•
	ISO 7816 - 8/9		•	•	•
Contactless Interface	ISO 14443 - 1/2/3/4	•	•	•	•
	ISO 14443 Type A	•	•	• (Default)	• (Default)
	ISO 14443 Type B			•	•
Common Criteria EAL6+		Chip Level (Combi/Contactless)	Chip Level	Chip Level	Chip Level
Global Platform 2.2.1				•	•
Java Card Classic 3.0.4				•	•
FIPS 140-2 (US)			FIPS Compliant	FIPS Compliant	
File Systems					
Transparent/Binary File		•	•	•	•
Linear Fixed Record		•	•	•	•
Linear Variable Record			•	•	•
Cyclic File			•	•	•
Cryptographic Capabilities					
3DES		56/112 bits	56/112/168 bits	56/112/168 bits	56/112/168 bits
AES			128/192/256 bits	128/192/256 bits	128/192/256 bits
RSA			Up to 4096 bits	Up to 4096 bits	Up to 2048 bits
ECC			Up to 521 bits	Up to 384 bits	Up to 544 bits
HASH			SHA1, SHA224, SHA256 SHA384, SHA512	SHA1, SHA224, SHA256 SHA384, SHA512	SHA1, SHA224, SHA256, SHA384, SHA512
CMAC			•	•	•
SCP02				•	•
Mutual Authentication		•	•	•	•
Secure Messaging		•	•	•	•
Random Number Generator		•	•	•	•
EEPROM Endurance					
100,000 Read/Write Cycles		Contact			
500,000 Read/Write Cycles		Combi/Contactless	•	•	•
Data Retention					
10 Years		•	•	•	•
30 Years			•	•	•

Subject to change without prior notice



ID DOCUMENT READERS

ACS's brand new product line of ID Document Readers delivers compact, versatile identity verification for space-constrained counters and kiosks. A modular design unifies OCR, RFID/NFC, and contact smart card reading to capture data quickly and reliably from ICAO Doc 9303–compliant e-passports and ID cards.



Physical Characteristics		
Dimensions (mm)	186,8 x 144,6 x 54,9	186,8 x 144,6 x 54,9
Weight (g)	358,0	372,0
Power		
USB Powered	●	●
Host Interface		
USB (Full Speed)	USB Type-C	USB Type-C
Expansion Port	●	●
E-passport Reading		
NFC Performance	≤3 seconds (chip data, ICAO 9303)	≤3 seconds (chip data, ICAO 9303)
MRZ Reading		
OCR Performance	≤1 sec, 99,8% accuracy	≤1 sec, 99,8% accuracy
Contactless Smart Card Interface		
ISO 14443 Type A and B	●	●
ISO/IEC 18092 (NFC)	●	●
ISO15693	●	●
MIFARE®	●	●
FeliCa	●	●
Smart Card Read Speed (kbps)	106, 212, 424, 848 (ISO 14443), 26 (ISO 15693)	106, 212, 424, 848 (ISO 14443), 26 (ISO 15693)
Reading Distance (mm) (depends on tag type)	up to 70	up to 70
NFC Reader/ writer Mode	●	●
NFC Card Emulation Mode	●	●
NFC Keyboard Emulation Mode	●	●
Contact Smart Card Interface		
ISO 7816 (Class A, B and C)	●	●
Smart Card Read/ Write Speed (kbps)	1250	1250
SAM Card Interface		
SAM Slot	1 (ISO 7816 Class A)	1 (ISO 7816 Class A)
Smart Card Read/ Write Speed (kbps)	1250	1250
Upgradeability		
USB Firmware Upgrade	●	●
QR Code & Barcode		
1-D		●
2-D		●
Built-In Peripherals		
LED	5	5
Buzzer	●	●
Certifications/Compliance		
PC/SC, CCID	●	●
ICAO DOC 9303	●	●
CE, FCC	●	●
RoHS, REACH	●	●
Supported OS Platforms		
Windows	●	●
Linux	●	●

Subject to change without prior notice



FIDO® DEVICES

ACS's FIDO® certified devices are compliant with FIDO U2F and FIDO2 protocols, and aim to provide users with a speedy and safeguarded passwordless online login experience.



Physical Characteristics				
Dimensions (mm)	29,0 × 14,5 × 10,5	ISO Card (Full Size)	35,0× 18,0 × 7,8	34,0 x 18,0x7,8
Weight (g)	4,5		4,2	4,2
Cryptographic Specifications				
ECC p256	●	●	●	●
Resident Key Storage	200	200	200	100
Interface				
USB (Full Speed)	USB Type-A		USB Type-C	USB Type-C
ISO 14443 Type A		●	●	●
ISO 7816 1/2/3		●		
Built-In Peripherals				
Button	1 with blue back-lit		1 with blue back-lit	
LED	1 (Blue)		1 (Blue)	1 (Blue)
Fingerprint Sensor				●
Certifications/Compliance				
FIDO U2F, FIDO2 L1	●	●	●	●
CE, FCC, UKCA	●		●	●
RoHS, REACH	●	●	●	●
Common Criteria EAL6+		Chip Level	Chip Level	Chip Level
Other Features				
Key Management Supported	●	●	●	●
Enterprise Attestation	Upon Request	Upon Request	Upon Request	Upon Request
Supported OS Platforms				
Windows	●	●	●	●
Linux	●	●	●	●
macOS	●	●	●	●
Android	●	●	●	●
iOS/iPad OS (iOS16 or above)	●	●	●	●
Browsers Support				
Chrome	●	●	●	●
Safari	●	●	●	●
Microsoft Edge	●	●	●	●
Firefox	●	●	●	●

Subject to change without prior notice



Reliable
Robust casing design



Strengthen Account Security
FIDO U2F, FIDO2, FIDO2 CTAP2 authenticators



Enable Swift Online Login
Passwordless operations



AUTOMATIC FARE COLLECTION TERMINALS

ACS develops and provides multi-interface devices to offer the convenience of cashless payment in transportation, parking and retail establishments such as fast food chains and convenience stores. These terminals support ISO 14443 Type A & B, MIFARE®, FeliCa, any of 4 types of NFC (ISO18092), Mastercard® Contactless or Visa payWave® cards/ tags, and printed or mobile barcodes. Accepting a wide range of payment options, this product line enables you to speed up operations and defend transactions against evolving threats.

ACR350



ACR350



COMING SOON

ACR360

Physical Characteristics

Dimensions (mm)	245.0 x 135.0 x 91.0 (Main Body) 245.0 x 135.0 x 159.0 (Main Body + Back mount)	278.0 x 110.0 x 92.0 (Main Body) 278.0 x 111.0 x 136.0 (Main Body + Back mount)
Weight (kg)	1.01 (Main Body) 1.81 (Main Body + Back mount)	TBD
Mounting	Pole Dimension: 31mm, 32mm, 35mm (Both Vertical and Horizontal)	Pole Dimension: 31mm, 32mm, 35mm (Both Vertical and Horizontal)

Processor

Processor	Arm® Cortex®-A9 Quad-Core 1GHz Processor	Arm® Cortex®-A55 Octa-Core 1.8GHz Processor
Embedded Platform	Linux	Android

Operation Modes

Standalone	●	●
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Power

Supply Voltage	9-36 VDC Reversed voltage polarity protection: up to -24 VDC Over voltage protection: up to +60 VDC	9-36 VDC Reversed voltage polarity protection: up to -24 VDC Over voltage protection: up to +60 VDC
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Power Rating (Max.)	48 W	48 W
SuperCap UPS	5 seconds with full function after primary power source is lost	5 seconds with full function after primary power source is lost

Contactless Smart Card Interface

ISO 14443 Type A and B	●	●
ISO/IEC 18092 (NFC)	●	●
MIFARE	●	●
FeliCa	●	●
Calypso	●	●
Smart Card Read/Write Speed (kbps)	106, 212, 424	106, 212, 424, 848
Reading Distance (mm)	up to 50	up to 50
NFC Reader/Writer Mode	●	●

Memory

Flash	25 GB	16 GB RAM (32 GB optional)
RAM	1 GB	2 GB
Micro SD Card Slot	Up to 128 GB	Up to 128 GB

Upgradeability

Firmware Upgradable	●	●
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Built-In Peripherals

SAM Slot	4 (Size: ID-000)	4 (Size: ID-000)
LCD Type	4.3" Color LCD	5.46" Color LCD
LCD Resolution	480 x 272	720 x 1280
Touch Panel	Capacitive Type	Capacitive Type
Button	4 with blue back-lit	2
LED	6 (4pcs at the front and 2pcs at the back)	6
Built-in Speaker	●	●
Buzzer	●	●
Tamper Switch	●	●
Real Time Clock	●	●
USB Port	USB Host (Type A)	USB 3.1 Type-A port x1 (5 Gbps)
Serial Port	RS-232 x 1, RS-485 x 1	RS-232 x 1, RS-485 x 1
Wi-Fi	2.4GHz, IEEE 802.11 b/g/n	2.4GHz, IEEE 802.11 b/g/n/ac
Network	4G/3G/GPRS	GPRS/GSM/3G/4G (LTE Cat 4)
SIM Slot	1 (Size: ID-000)	1 (Size: ID-000)
Ethernet	10/100/1000 Mbps	10/100/1000 Mbps with POE support
Positioning System	GPS, Glonass, Beidou, Galileo AGPS Support	GPS, Glonass, Beidou, Galileo AGPS Support
Bluetooth®	5.1 Dual Mode	5.0 BLE
Control Signals	2 x Input + 2 x Output	2 x Input + 2 x Output
Ignition Sense	●	●
CAN Bus	●	●
1D/2D Barcode Scanner	●	●

Other Features

Key Management Supported	Fixed Key, MK/SK, DUKPT	DUKPT
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Certifications/Compliance

EMV Level 1 & Level 2 (Contactless)	●	●
PCI	●	●
MasterCard® Contactless,	●	●
Visa payWave®	●	●
AMEX, Discover, UnionPay, JCB	●	●
Apple pay , Google Pay	●	●
Apple VAS, Google Smart Tap	●	●
CE, FCC	●	●
RoHS, REACH	●	●
Shock and Vibration	Military Standard MIL-STD-810	Military Standard MIL-STD-810
IP Rating	IP65	IP65
IK Rating	IK08	IK08

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